Synthetic controller protocol

1. Shared wire definitions

QA-TABLE-20261030. The simulator is the controller. Every multibyte wire integer is little endian. Fields are serialized in table order with no padding, scaling, checksum or trailing bytes.

Header field	Wire type	Automatic behavior
Id	uint8	role=id
Size	uint8	role=length; total frame bytes, including this three-byte header
Seq	uint8	role=seq; max=255; scope=link; response=count

Enum Toggle has exactly Off=0 and On=1. Scalar fields do not have a count attribute. Only the Limits array on page 4 has count=2. No Types/Struct declarations are required.

Message inventory

2. Requested physical pages 2-4

Define both listed messages. Follow shared header and Toggle references to physical page 1. Directions in this table describe the physical device, not automatically the simulator.

Code / name	ID	Device direction	Total bytes / detail
SENSOR_STATUS Sensor status	0x91	Device transmits to controller	6 page 3
SET_LIMITS Set limits	0x92	Device receives from controller	7 page 4

Repeated column headings are not fields. Only the two inventory rows are messages.

SENSOR_STATUS

3. Body fields, in wire order

Field	Wire type	Count	Meaning / reference
Value	uint16	scalar	Unscaled sensor reading
State	uint8	scalar	Enum Toggle, see shared definition on page 1

Header is not repeated in this body. Body length is 3 bytes; total length is 6 bytes. Use canonical names exactly: SENSOR_STATUS, Sensor status, Value, State.

SET_LIMITS

4. Body fields, in wire order

Field	Wire type	Count	Meaning / reference
Limits	int16	2	Two signed fixed-array values. Index 0 is serialized before index 1.

Header is not repeated in this body. Body length is 4 bytes; total length is 7 bytes. Use canonical names exactly: SET_LIMITS, Set limits, Limits. No additional messages are in scope.